

| Question | Answer                                                                                                                                                                                                                                      | Marks     | Guidance                                                                                                                                                                                                                                                                                                                                               |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4(a)(i)  | magnitudes (of electric fields) are equal                                                                                                                                                                                                   | <b>B1</b> | <b>Allow</b> 'strength' for 'magnitude'                                                                                                                                                                                                                                                                                                                |
|          | <u>Any two from:</u> <ul style="list-style-type: none"> <li>(electric field) due to proton is to the right</li> <li>(electric field) due to electron is to the right</li> <li>directions (of electric fields) are the same</li> </ul>       | <b>B2</b> | <b>Allow</b> towards electron / away from proton<br><b>Allow</b> towards electron / away from proton.<br><b>Do not allow</b> if both directions given to the left.<br><br>If all three points are attempted, and two are correct and one is incorrect, treat as one contradiction and award 1/2 (CON).                                                 |
| 4(a)(ii) | magnitudes (of electric potentials) are equal                                                                                                                                                                                               | <b>B1</b> |                                                                                                                                                                                                                                                                                                                                                        |
|          | <u>Any two from:</u> <ul style="list-style-type: none"> <li>(electric potential) due to proton is positive</li> <li>(electric potential) due to electron is negative</li> <li><u>signs</u> (of electric potentials) are opposite</li> </ul> | <b>B2</b> | <b>Do not allow</b> if the signs are the wrong way around.<br><b>Do not allow</b> 'in opposite directions'<br><br>If all three points are attempted, and two are correct and one is incorrect, treat as one contradiction and award 1/2 (CON).<br>If BP3 is incorrect purely due to use of 'in opposite directions', this does not negate BP1 and BP2. |
| 4(b)(i)  | $F = Q_1 Q_2 / 4\pi\epsilon_0 R^2$                                                                                                                                                                                                          | <b>C1</b> | C1 mark is for any correct general statement of Coulomb's law, using standard symbols.<br><b>Allow</b> $Q$ , $q$ and $e$ for $Q_1$ and $Q_2$ .<br><b>Allow</b> $x$ or $r$ for $R$ .                                                                                                                                                                    |
|          | $F = e^2 / 16\pi\epsilon_0 x^2$                                                                                                                                                                                                             | <b>A1</b> | Treat correct answer as AFC.<br><b>Ignore</b> sign                                                                                                                                                                                                                                                                                                     |
| 4(b)(ii) | $E_P = 2Fx$                                                                                                                                                                                                                                 | <b>A1</b> | <b>Ignore</b> sign                                                                                                                                                                                                                                                                                                                                     |